Millend Roy

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Education

Columbia University *New York, USA*
Ph.D. in Operations Research (CGPA : 4.14/4.00) *Sep 2023 - May 2028*
M.S. in Operations Research (CGPA : 3.75/4.00) *Sep 2023 - Aug 2024*
Advisor : Prof. Agostino Capponi

Indian Institute of Technology - IIT, Dhanbad *Dhanbad, India*
B.Tech in Electrical Engineering (CGPA : 9.51/10, *Institute Silver Medalist*) *Jul 2017 - May 2021*
Advisor : Dr. Bhukya Krishna Naik

Work Experience

BP America, Inc., *Quantitative Strategist Intern, FTR Trading in Power Markets* *Jun 2026 - Aug 2026*
Scale AI, Inc., *GenAI Specialist, Human Frontier Collective Intern* *May 2025 - Aug 2025*
Microsoft Research Lab India, *Societal impact through Cloud & AI Research Fellow* *Feb 2021 - Jul 2023*
Hurrey Tech Ventures R&D, *Data Science*: led team to SAP ET Innovation Awards'20 *Apr 2020 - Sep 2020*

Research Projects

FFT-Accelerated Field Integration on Trees via Backbone Decomposition. *Jan 2026 - Sep 2026*
Under Supervision of Prof. Krzysztof Choromanski (Google DeepMind Robotics & Columbia University)

- Formulated scalable tree field integration for graph-structured kernels, targeting efficient computation of distance-based vertex interactions that arise in *graph kernels*, *mesh interpolation*, and *masked Transformer attention*.
- Developed a backbone decomposition algorithm using path/spider-chain structure, anchor-depth aggregation, and one-/two-dimensional FFT-based Toeplitz and Hankel convolutions to accelerate exact tree-kernel aggregation.
- Proved $O(n \log n)$ runtime for key tree families and designed downstream evaluations including tree-based Relative Positional Encoding (RPE) masks for Vision Transformers, graph classification with learned rational f-kernels, and mesh/point-cloud interpolation benchmarks.

PREFER: Aligning LLM-Generated Summaries with Evolving User Preferences. *Oct 2025 - May 2026*
Under Supervision of Prof. Agostino Capponi and Prof. Vineet Goyal (Columbia University)

- Formulated personalized review summarization as a human-centered AI alignment problem, to address (i) user-information overload and the limitations of (ii) one-size-fits-all product review summaries.
- Developed an interpretable, aspect-based preference representation by processing *583K+* reviews and *1.33M+* sentences using sentence embeddings, PCA, clustering, and soft aspect assignments.
- Built PREFER, a feedback-driven alignment framework combining (i) personalized evidence selection, (ii) LLM-based abstractive rewriting, and (iii) online preference updates via entropic mirror descent.
- Evaluated on Amazon Reviews'23 across cross-user, stationary, and drifting-preference settings, achieving up to 99.95% preference-evidence alignment and decreasing surrogate regret across seeds.

CapOptix: An Options-Framework for Capacity Market Pricing Models. *Apr 2024 - Oct 2025*
Graduate Research Assistant with Prof. Agostino Capponi and Prof. Vijay Modi (Columbia University)

- Studied *"missing money"* problem in electricity markets where high-capital, low-variable-cost generators (e.g., renewables) struggle to recover fixed costs through volatile spot prices.
- Developed a regime-switching model for electricity prices and *derived option premiums* using *Monte-Carlo* simulations under multiple stochastic regimes; embedded *CVaR-based risk metrics* for decision-making under price-uncertainty.
- Conducted experiments for New York & German energy mixes (having different renewable penetration rates).
- Results show a *10-15% improvement* in cost recovery when compared with existing capacity remuneration mechanisms.

EnCortex: A framework for decision management in Energy Systems. *Jul 2021 - Jul 2023*
Project:Vasudha with Dr. Akshay Nambi and Dr. Shivkumar Kalyanaraman (Microsoft Research India)

- Built core easily-extensible energy *abstractions* for: *Price Arbitrage, Portfolio Optimization and Demand Bidding.*
- Paired with modular optimizers including *stochastic, mixed integer linear programming and reinforcement learning* techniques enhancing developer productivity and surpassing state-of-the-art benchmarks by 5-6%.
- *MLOps* for fine-tuning RL models: *Behaviour Cloning, OfflineRL and Transfer Learning*, with efficiency gains of 3%.

Responsibilities: Teaching, Reviewing, and Presentations

Ph.D. Enhanced Teaching Assistant at **Columbia University** for :

- Statistical Analysis and Time Series (IEOR 4709) by Prof. Agostino Capponi. *Jan 2026 - May 2026*
- Algorithmic Trading (IEOR 4733) by Prof. Christopher Perez. *Jan 2025 - May 2025*
- Applications Programming for Financial Engineering (IEOR 4500) by Dr. Anran Hu. *Sep 2024 - Dec 2024*
- Analysis of Algorithms (CSOR 4231) by Dr. Rachel Cummings. *Jan 2024 - May 2024*

Reviewer: WWW'26, KDD'26, NeurIPS'26, IEEE Transactions on Energy Markets Policy and Regulation

INFORMS Annual Meeting'26, San Francisco, CA, USA. (Invited Speaker) *Nov 2026*
INFORMS Annual Meeting'25 PG&E Analytics Award Session, Atlanta, GA, USA. *Oct 2025*
Future Energy Summit'25, Boston, MA, USA. (Invited Speaker) *Feb 2025*

Selected Publications

Preprints/ Working Paper * - equal contribution

3. *Structure-Adpative Tree Field Integrators.* **Millend Roy**, Soham Samal, Ivan Zelich, Krzysztof Choromanski. Submitted, 2026.
2. *PREFER: Personalized Review Summarization with Online Preference Learning.* **Millend Roy**, Agostino Capponi, Vineet Goyal. Submitted, 2026.
1. *CapOptix: An Options-Approach based Framework for Capacity Market Pricing Models.* **Millend Roy**, Agostino Capponi, Vladimir Pyltsov, Yinbo Hu, Vijay Modi. Revising, 2025.

Journal

2. Machine Learning Based Adaptive Fault Diagnosis in Active Distribution Network. Sourav Kumar Sahu, **Millend Roy**, Soham Dutta, Dusmanta Kumar Mahanta. In: *Electric Power Systems Research EPSR'23.*
1. A Data-Driven Fault Detection Approach using Smart Meters in Modern Distribution System. Soham Dutta*, Sourav Kumar Sahu*, **Millend Roy**, Pradip Kumar Sadhu. In: *Sustainable Energy, Grids and Networks SEGN'23.*

Conference

4. AlphaSearch: Agentic AI for Price Arbitrage in Energy Systems. **Millend Roy***, Vladimir Pyltsov*. In: *32nd SIGKDD Conference 2026 on Knowledge Discovery and Data Mining, KDD'26 AI4Sciences, Jeju, Korea.*
3. Dynamic Incentive Design for Demand Response in MicroGrids. **Millend Roy**, Vladimir Pyltsov, Ruturaj Godse, Tony Dear. In: *2025 16th ACM Conference on Sustainable Energy Systems, E-ENERGY '25, Netherlands.*
2. Reliable Energy Consumption Modeling for an Electric Vehicle Fleet. **Millend Roy**, Akshay Nambi, Anupam Sobti, Tanuja Ganu, Shivkumar Kalyanaraman, Shankar Akella, Jaya Subha Devi, and S A Sundaresan. In: *2022 5th ACM Conference on Computing and Sustainable Societies. COMPASS'22. Seattle, WA, USA.*
1. Renewable Energy and Demand Forecasting in a Smart Grid. Vishnu Vardhan Sai Lanka*, **Millend Roy***, Shikhar Suman, Shivam Prajapati. In: *2021 IEEE Innovations in Energy Management, IEMRE'21. Kolkata, India.*

Skills

Languages	Python, C++, C, Matlab, Simulink, Tableau, Docker, Git, Dataiku.
Frameworks	PyTorch, CrewAI, AgenticsIBM, StableBaselines, RLlib, Rsome, Scikit-Learn, Keras, NumPy, Pandas, SciPy, Matplotlib, Seaborn, Streamlit, Gurobi, Google OR-Tools, Cvxpy.
Courses	Agentic AI, Machine Learning, Algorithmic Game Theory, Computational Reinforcement Learning, Linear, Convex and Combinatorial Optimization, Stochastic Processes, Analysis of Algorithms.

Select Awards and Honors

- Selected for **JPMorganChase Quantitative Trading and Research** Mentorship Program'26. *May 2026*
- **Winner** of *GridTech Hackathon*'26, sponsored by NY Power Authority and National Grid. *Apr 2026*
- **Ranked** 2nd in the *PG&E Energy Analytics Challenge* by IISE ESD and INFORMS QSR Sections. *Jun 2025*
- Recipient of **Columbia Piyasombatkul Fellowship** at Columbia Engineering. *Sep 2023*
- *ASEAN India Hackathon*'21 **Global Finalist**. *Jan 2021*
- **Winner** of the Grand Finale of *Smart India Hackathon*'20. *2020*
- *Samsung Innovation Awards* **Finalists**. *2020*
- SURGE'20 opportunity at IIT Kanpur. 2020
- Recipient of **Science Academies' Summer Fellowship** from IASc-INSA-NASI, India. *2019*